Lesson plan · for the teacher

Protecting models and data: unit check

AI security · Level 3 · Unit check

Time	Materials
45 min	Board, chalk, paper. No devices needed.

Learning goals

- Poisoned data
- Models that remember too much
- Pictures that fool a model
- Is this the real model?

Activity: Team quiz

1. Make teams of four
2. Read each question aloud; teams agree on one answer
3. Reveal the answer and ask one team to explain why
4. Write the hardest question on the board for tomorrow

Discussion

- Which check will you start with at your school?
- Which question surprised you, and why?
- How would you explain a fingerprint to a colleague?

Exit ticket

- One check I will start using, and where

Try it in class tomorrow

Pick the check from this unit your school needs most (median checks on class data, redacting before sharing, a person deciding what matters, or a list of AI tools with their sources), start it this week, and tell a colleague how it went.

Worksheet · for pupils

Protecting models and data: unit check

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Name _____ Class _____ Date _____

1 **What is data poisoning?**

- ◯ Slipping wrong or planted examples into training data
- ◯ Breaking into the computer that runs the model
- ◯ Asking a model too many questions at once

2 **A backdoor makes a model misbehave only when a hidden trigger appears, so normal testing rarely finds it.**

- ◯ True
- ◯ False

3 **What does this program print?**

```
data = [50, 52, 49, 51, 5]
m = sorted(data)[len(data) // 2]
for x in data:
    if abs(x - m) > m / 2:
        print("check", x)
```

- ◯ check 5
- ◯ check 52
- ◯ Nothing

4 **A school wants to train a homework helper on pupils' past work. Which steps protect pupils' privacy? Choose all that apply.**

- ☐ Leave out work the helper doesn't need
- ☐ Remove names, numbers and rare details first
- ☐ Test with a planted canary before anyone uses it
- ☐ Train on everything, then delete the original files

5 **Match each risk to a defence.**

Poisoned training data	• • • •	Check the fingerprint and signature
A model that memorises private details	• • • •	Test with small changes; a person decides
Adversarial examples	• • • •	Keep personal data out of training
A tampered or fake model file	• • • •	Know the source; keep a clean test set